# Raj Inder Singh Gulati

Vellore, India | rajindersinghgulati01@gmail.com | +91 9515513245
LinkedIn | GitHub | Medium | LeetCode

## SUMMARY

Computer Science student (CGPA: 9.03/10) at VIT Vellore with 2 internships and 4+ shipped projects in full-stack development, AI/ML systems, and cybersecurity. Published researcher in federated learning who has delivered agentic AI platforms, multi-node inference pipelines, and container security tooling. Proficient in Java, Python, Go, C++, Spring Boot and React.

## EDUCATION

**B.Tech — Computer Science & Engineering** 2023–2027
VIT University, Vellore CGPA: 9.03/10

## PROFESSIONAL EXPERIENCE

**AI Engineering Intern** May – Jun 2026
*EigenAxis.ai*

- Designed the Agentic Tools framework and multi-agent orchestration layer for **LLM Plus**, an AI platform supporting multiple pluggable SLM backends and serving enterprise workloads in production.
- Engineered a FastAPI-based orchestration backend exposing REST endpoints (chat, health, configuration), reducing agent response latency by 30% through asynchronous request handling.
- Implemented the full agentic pipeline—client session management, multi-format response serialization (JSON/XML/YAML), and a request routing engine coordinating 3+ tool-calling agents per request.
- Configured Distributed Data Parallel (DDP) inference pipelines for decoder-only language models using `torchrun`, achieving 2.4× throughput improvement across multi-GPU setups.
- Managed code reviews via Gerrit, maintained 15+ YAML-driven model configurations, and shipped the frontend web interface enabling live agent interaction for internal stakeholders.

**Web Developer Intern** Jun – Oct 2025
*KalkiNI*

- Developed letsgetmentor.com from concept to launch, delivering a mentorship platform that onboarded 100+ users within the first month.
- Led a 4-person back-end team from July–October 2025, owning AWS deployment, CI/CD pipeline setup, and infrastructure management that maintained 99.5% uptime.

## PROJECTS

**FedGuard — Federated Intrusion Detection System** 2026
*Python, PyTorch, Flower (flwr), Spring Boot, React, Recharts*

- Created a federated network security system where 5 decentralized clients train locally on 50,000+ traffic samples without sharing raw data with the central server.
- Orchestrated a federated learning pipeline using the Flower framework; federated clients collaboratively train anomaly detection models, achieving 94% detection accuracy via FedAvg aggregation.
- Built a Spring Boot REST backend to coordinate FL rounds, aggregate model weights, and serve threat predictions through a 6-endpoint API layer with sub-200ms response times.
- Delivered a React dashboard featuring 4 live-polling visualizations (attack rate trends, alert timeline, client health status, confidence distribution) using Recharts and Axios.
- Integrated confidence-scored inference classifying traffic as attack or benign with 0% raw data exposure, reducing false-positive alerts by 40% compared to single-scanner baselines.

**CSVA — Consensus-Based Container Vulnerability Scanner** 2026

*Go, Trivy, Grype, CLI, JSON Schema Design*

- Constructed a Go CLI tool that concurrently executes 2+ industry-standard scanners (Trivy, Grype) against container images and synthesizes results through a weighted consensus engine.
- Designed an interface-based scanner adapter pattern with a unified vulnerability schema, enabling integration of new scanners in under 50 lines of configuration.
- Computed agreement percentages, per-scanner confidence scores, and unique findings—consolidating contradictory reports into a single ranked vulnerability view.
- Generated structured reports separating confirmed vulnerabilities (agreed by $\geq$2 scanners) from disputed findings, cutting false-positive triage time by 60% for DevSecOps teams.

**Apartment Management App — MERN Stack** 2025

*MongoDB, Express.js, React, Node.js, JWT*

- Launched a full-stack apartment management platform with role-based access control (tenant/admin), 12+ RESTful API endpoints, JWT authentication, and MongoDB storage.
- Developed a responsive React frontend supporting CRUD operations across 6 modules (maintenance, billing, announcements, residents, complaints, amenities) with centralized error handling.

## RESEARCH & PUBLICATIONS

- **Federated Modality Aware Network for Robust Glioblastoma Survival Prediction** — Accepted at conference. Proposed a federated learning framework for multimodal MRI-based glioblastoma survival prediction; trained across 3 simulated institutions using pFedAvg, gated cross-attention fusion, and conformal calibration, achieving 89% concordance index.
- **Quantum Birthday Attack Prevention via KECCAK-512** — Co-authored a theoretical analysis demonstrating that KECCAK-512 sponge construction (SHA-3 XOF) raises quantum collision attack complexity from $O(2^{n/3})$ to $O(2^{n/2})$, restoring classical security margins.
- Published technical articles on Medium.

## ACHIEVEMENTS

- Solved 500+ algorithmic problems on LeetCode.
- Competed in 4+ campus hackathons and inter-collegiate tech events at VIT.

## TECHNICAL SKILLS

**Languages:** Java, C++, Python, Go, C, SQL, JavaScript

**Backend & Frameworks:** Spring Boot, FastAPI, Node.js, Express.js, REST APIs, JWT, Docker

**Frontend:** React, HTML/CSS, Recharts, Axios

**Databases:** PostgreSQL, MongoDB, MySQL, H2DB

**AI/ML:** PyTorch, Scikit-learn, Federated Learning (Flower), Distributed Data Parallel, LLM Orchestration

**Security:** Kali Linux, Nmap, Burp Suite, Wireshark, Metasploit, OWASP Top 10, Aircrack-ng, Bettercap, BeEF, Hashcat

**Cryptography:** RSA, ECC, KECCAK-512, Hash Functions, OpenSSL, JCrypTool

**DevOps & Tools:** Git, GitHub, Docker, Google Colab, Gerrit, AWS, CI/CD, Cisco Packet Tracer